Ahmed Ghaffar Packages.

Date:

outlook	Temp	Humidity	windy	Rain
OL	T	H	W	R
Sunny	Hot	High	weak	No
Sunny	Mild	High	Strong	No
Cloudy	Hot	High	weak	Yes
Rainy	Mild	high	weak	Yes
Rainy	Cool	Normal	weak	Yes
Rainy	Cool	Normal	Strong	No
Cloudy	Mild	Normal	Strong	Yes
Sunny	Cool	Normal	weak	No

Root Node Identification

Step: 1 Calculate the entropy of the target Rain

Total Records = 8

Yes = 4

No = 4

Entropy:

$$H(S) = -\sum_{i} p_i \log_2 (p_i)$$

$$H(S) = -\frac{4}{8} \log_2 \frac{4}{8} - \frac{4}{8} \log_2 \frac{4}{8}$$

Yes Probability:

$$P(Yes) = \frac{\text{Yes Records}}{\text{Total Records}} = \frac{4}{8} = 0.5$$

No Probability

$$P(No) = \frac{\text{No Records}}{\text{Total}} = \frac{4}{8} = 0.5$$

CamScanner

Ahmed Ghaffar Packages.

Date:

C. Info. Gain of Humidity:

1. Humidity	Total	Y	N
High	4	2	2
Normal	4	2	2

2. Entropy

$H(\text{High}) = -\left[\frac{2}{4}\log_2\frac{2}{4} + \frac{2}{4}\log_2\frac{2}{4}\right]$

$= 1.000$

$H(\text{Normal}) = -\left[\frac{2}{4}\log_2\frac{2}{4} + \frac{2}{4}\log_2\frac{2}{4}\right]$

$= 1.000$

3. Weighted entropy:

$H(S/\text{Hu}) = \frac{4}{8}(1.000) + \frac{4}{8}(1.000)$

$= 1.000$

4. Info gain of Temp

$H(S/\text{H}) = 1.000 - 1.000$

$= 0$

Infinix HOT 11

CamScanner

4. Info gain of OL:

IG (OL) = 1.000 − 0.344

= 0.656

B. Info Gain of Temperature (T):

1. Temperature	Total	Yes	No
Hot	2	1	1
Mild	3	2	1
Cool	3	1	2

2. Entropy

$H(\text{hot}) = -\left[\frac{1}{2}\log_2\frac{1}{2} + \frac{1}{2}\log_2\frac{1}{2}\right] = 1.000$

$H(\text{Mild}) = -\left[\frac{2}{3}\log_2\frac{2}{3} + \frac{1}{3}\log_2\frac{1}{3}\right] = 0.918$

$H(\text{Cool}) = -\left[\frac{1}{3}\log_2\frac{1}{3} + \frac{2}{3}\log_2\frac{2}{3}\right] = 0.918$

3. weighted entropy

$H(S/T) = \frac{2}{8}(1.000) + \frac{3}{8}(0.918) + \frac{3}{8}(0.918)$

$= 0.9385$

4. Info gain of ~~Age~~: Temp:

IG (Temp) = 1.000 − 0.9385

= 0.0615

Ahmed Ghaffar Packages.

Date:

D. Info gain of windy

1. windy

windy	Total	Y	N
weak	5	3	2
Strong	3	1	2

2. Entropy:

$$H(\text{weak}) = -\left[\frac{3}{5}\log_2\frac{3}{5} + \frac{2}{5}\log_2\frac{2}{5}\right]$$

$$= 0.970$$

$$H(\text{Strong}) = -\left[\frac{1}{3}\log_2\frac{1}{3} + \frac{2}{3}\log_2\frac{2}{3}\right]$$

$$= 0.918$$

3. weighted entropy

$$H_{\text{windy}} = \frac{5}{8}(0.970) + \frac{3}{8}(0.918)$$

$$= 0.9505$$

4. IG of windy

$$= 1.000 - 0.9505$$

$$= 0.049$$

Root Node is Temperature

Ahmed Ghaffar Packages.

Date:

$$H(S) = -[0.5 \log_2(0.5) + 0.5 \log_2(0.5)]$$
$$= -[0.5(-1) + 0.5(-1)]$$
$$H(S) = 1.000$$

Step:2 Info Gain for each attribute

1. Info Gain of outlook (OL):

1. outlook	Total	Yes	No
Sunny	3	0	3
Rainy	3	2	1
Cloudy	2	2	0

2. Entropy

$H(\text{sunny}) = 0$

If all the records are from one class in the entropy will be '0'

$$H(\text{Rainy}) = -\left[\frac{2}{3}\log_2\frac{2}{3} + \frac{1}{3}\log_2\frac{1}{3}\right]$$

$$= 0.918$$

$H(\text{cloudy}) = 0$.

3. weighted entropy:

$$H(S/OL) = \frac{3}{8}(0) + \frac{3}{8}(0.918) + \frac{2}{8}(0)$$

$$H(S/OL) = 0.344$$

Tinix HOT 10i

CamScanner